



How big is the physical activity intention–behaviour gap? A meta-analysis using the action control framework

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Objectives. The physical activity (PA) intention–behaviour gap is a topic of considerable contemporary research, given that most of our models used to understand physical activity suggest that intention is the proximal antecedent of behavioural enactment. The purpose of this study was to quantify the intention–PA gap at public health guidelines with a meta-analysis of the action control framework.

Design. Systematic review and meta-analysis.

Methods. Literature searches were conducted in July 2012 among five key search engines. This search yielded a total of 2,865 potentially relevant records; of these, 10 studies fulfilled the full eligibility criteria ($N = 3,899$).

Results. Random-effects meta-analysis procedures with correction for sampling bias were employed in the analysis for estimates of non-intenders who subsequently did not engage in physical activity (21%), non-intenders who subsequently performed physical activity (2%), intenders who were not successful at following through with their PA (36%), and successful intenders (42%). The overall intention–PA gap was 46%.

Conclusion. These results emphasize the weakness in early intention models for understanding PA and suggest this would be a problem during intervention. Contemporary research that is validating and exploring additional constructs (e.g., self-regulation, automaticity) that augment intention or improving the measurement of motivation seems warranted.

Statement of contribution

What is already known on this subject? Intention is considered the proximal antecedent of behaviour in many popular models. Intention is also an established correlate of physical activity behaviour, yet discordance is considerable in experimental research.

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What does this study add? This meta-analysis of studies that have assessed concordance/discordance of physical activity intention and behaviour at public health guidelines shows the intention-behaviour gap at 48% and the discordance is from intenders who do not act. The results demonstrate that discordance is not just from extreme levels of intention or behaviour (e.g., intend to exercise six times but only exercise five), but from levels that are relevant to health promotion.

Many theories applied to physical activity include intention, the motivation to enact a behaviour, as the proximal antecedent to behavioural performance (Bandura, 2004; Fishbein *et al.*, 2001; Noar & Zimmerman, 2005). Overall, this construct has been validated as the dominant predictor of physical activity in correlational tests of the theory of planned behaviour (Symons Downs & Hausenblas, 2005), protection motivation theory (Plotnikoff & Trinh, 2010), and variants of social cognitive theory (Roberts, Maddison, Magnusson, & Prapavessis, 2010) and self-determination theory (Hagger & Chatzisarantis, 2009). Intention is also the pivotal construct in stage-based algorithms such as the stages of change used in the transtheoretical model (Nigg *et al.*, 2011). Meta-analyses have placed the point estimate of the intention-behaviour relationship as $r = .50$ (Armitage & Conner, 2001; Hagger, Chatzisarantis, & Biddle, 2002; McEachan, Conner, Taylor, & Lawton, 2011; Symons Downs & Hausenblas, 2005), which situates it within the large effect size range (Cohen, 1992) and larger than any other known correlate of physical activity.

Despite this strong predictive value of intention in the physical activity domain, tests of its absolute predictive utility show discordance with behaviour. For example, a recent meta-analytic examination of experimental changes in intention-behaviour relationships showed that medium-sized changes in intention resulted in trivial-sized changes in behaviour ($r = .06$) (Rhodes & Dickau, 2012a). This line of enquiry has generated the term 'intention-behaviour gap' in the scientific community.

Given the prominence of the intention construct in health behaviour models, intention-behaviour discordance has sparked researchers to attention. Some researchers have produced and subsequently validated models that postulate post-motivational planning and self-regulation constructs meant to bridge the intention-behaviour relationship (Gollwitzer, 1999; Schwarzer, 2008; Sniehotka, 2009). Dual-process models suggest and provide evidence that automaticity/habit may also account for some of this discordance (Gardner, de Bruijn, & Lally, 2011).

Clearly, many variables have been posited to account for intention-behaviour discordance (Rhodes & Dickau, 2012b). Still, it seems helpful to appraise the severity of intention-behaviour discordance within the context of meeting physical activity guidelines. Deviations in intention-behaviour relations at very high points on the curve (e.g., intending to exercise seven times per week but achieving only five times per week) or an extremely low point (intending to do a bout of exercise but doing nothing) clearly have low value when applied to physical activity promotion. Most interventions are designed to promote physical activity at public health guidelines, so this would appear the most important area to understand the severity of the intention-behaviour gap (Foster, Hillsdon, & Thorogood, 2009). The action control framework may assist in the process (Rhodes, Courneya, & Jones, 2003). In the action control framework, intention and subsequent behaviour are divided into quadrants by the criterion of physical activity at public health guidelines. This creates four possible quadrants for the intention-behaviour relationship, including two concordant (non-intenders who are subsequently not active; successful intenders) and two discordant (non-intenders who are subsequently active; unsuccessful intenders) profiles (see also Godin, Shephard, & Colantonio, 1986; Sheeran,

2002 for alternative labels). An understanding of intention–behaviour discordance using this framework would provide evidence for whether intention-based theories need augmenting in interventions.

Thus, the purpose of this study was to overview the magnitude of the intention–behaviour profiles in the physical activity domain using meta-analysis in order to ascertain the size of the physical activity intention–behaviour gap when considering public health guidelines.

Method

Eligibility criteria

A study was included in this review if it met the following inclusion criteria: (1) a measure of leisure-time PA (inclusive of exercise) as the dependent variable, (2) an assessment of physical activity intention, and (3) employment of the intention–behaviour relationship in any capacity where it was divided into quadrants (intention high/low; behaviour high/low). Excluded studies were those that were written in any language other than English.

Search strategy

A prior systematic review on intention–behaviour moderation in the physical activity domain served as the foundation for this search (Rhodes & Dickau, 2012b). This search was conducted for literature up to March 2010. A subsequent review update was performed from March 2010 until September 2012 using Academic Search Complete, ERIC, MEDLINE, PsycINFO, and SPORTDiscus. The electronic search strategy was developed by the authors of the prior meta-analysis (Rhodes & Dickau, 2012b). Combinations of the following keywords were used: intention, action control, inclined actor, inclined abstainer, disinclined actor, disinclined abstainer, goal, physical activity, exercise, walking, running, cycling, active transport, theory of planned behaviour, theory of reasoned action, health belief model, protection motivation theory, and social cognitive theory (see Appendix A). The search was restricted to prospective designs, but not limited by sample population. Manual cross-referencing of bibliographies was also completed. Finally, authors of papers found in the search were contacted and asked to provide any additional published work on the topic that might have been missed by the search.

Screening

Citations were screened by two reviewers (R.R. and H.M.) using pre-defined inclusion criteria and were initially screened based on the title, abstract, and full-text search of identified keywords. Consensus was 100% in this procedure. Relevant studies were then selected for a full read of the article. Potential studies for adjudication were examined independently by two reviewers (R.R. and HM). It was then determined whether the study met the criteria and was included in the review. Consensus was reached in 100% of the cases.

Data abstraction

Data were extracted from the studies using 10 pre-specified items by two reviewers (RR and GdB). The reviewers had complete agreement with their abstraction forms. These included the author, sample, design, independent measure, dependent measure, intention–behaviour profile, and outcomes (see Table 1).

Table 1. Studies employing the action control framework

Authors	Sample and design	Regular physical activity definition	Intention-behaviour profile formulation	Profile results
Godin <i>et al.</i> (1986)	163 university employees in a two-month prospective design	Vigorous exercise performed at least 2 × per week	Intention was measured with a 7-point Likert scale and high/low was dichotomized by yes representing the top 3 response categories; behaviour was measured as an open-ended frequency (GLTEQ). Exercise was defined as vigorous intensity, at least. Profiles were dichotomized at 2+ times per week.	Non-intenders (26%), non-intenders exceeding intentions (1%), unsuccessful intenders (28%), and successful intenders (45%)
Rhodes <i>et al.</i> (2003)	300 undergraduate students in a 2-week prospective design	Vigorous exercise performed at least 20 min per bout at least 3 × per week	Intention was measured with an open-ended frequency; behaviour was measured as an open-ended frequency (GLTEQ). Exercise was defined as vigorous intensity, at least 20+ min per bout. Profiles were dichotomized at 3+ times per week.	Non-intenders (33%), non-intenders exceeding intentions (2%), unsuccessful intenders (27%; mean bouts = 1.54, SD = 1.22), and successful intenders (38%)
Rhodes and Plotnikoff (2006)	1,192 Community-dwelling adults in a 6-month prospective design	Vigorous exercise performed at least 20 min per bout at least 3 × per week	Intention was measured as a single-item yes/no format; behaviour was measured as a single-item yes/no format. Physical activity was defined as vigorous intensity, 3 times per week for 20+ min per bout. Profiles were dichotomized around yes/no responses	Non-intenders (9%), non-intenders exceeding intentions (2%), unsuccessful intenders (43%), and successful intenders (46%)

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Table 1. (Continued)

Authors	Sample and design	Regular physical activity definition	Intention-behaviour profile formulation	Profile results
Rhodes <i>et al.</i> (2008)	611 adults of a workplace sample in a 6-month prospective design	Moderate- or vigorous -intensity physical activity performed at least 30 min per bout at least 4 × per week	Intention was measured as a single-item yes/no format; behaviour was measured as a single-item yes/no format. Physical activity was defined as moderate or vigorous intensity, 4+ times per week, at least 30 min per bout. Profiles were dichotomized around yes/no responses	Non-intenders (6%), non-intenders exceeding intentions (2%), unsuccessful intenders (35%), and successful intenders (57%)
Rhodes <i>et al.</i> (2010)	153 undergraduate students in a 2-week prospective design	Moderate- or vigorous -intensity physical activity performed at least 30 min per bout at least 4 × per week	Intention was measured with an open-ended frequency; behaviour was measured as an open-ended frequency (GLTEQ). Exercise was defined as moderate or vigorous intensity, at least 30+ min per bout. Profiles were dichotomized at 4+ times per week.	Non-intenders (35%), non-intenders exceeding intentions (7%), unsuccessful intenders (21%; mean bouts = 1.69, <i>SD</i> = 1.26), and successful intenders (37%)
Rhodes, Fiala, and Nasuti (2012)	216 undergraduate students in a 2-week prospective design	Vigorous exercise performed at least 20 min per bout at least 3 × per week	Intention was measured with an open-ended frequency; behaviour was measured as an open-ended frequency (GLTEQ). Exercise was defined as vigorous intensity, at least 20+ min per bout. Profiles were dichotomized at 3+ times per week.	Non-intenders (14%), non-intenders exceeding intentions (1%), unsuccessful intenders (36%; mean bouts = 1.27, <i>SD</i> = .80), and successful intenders (49%)

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Table 1. (Continued)

Authors	Sample and design	Regular physical activity definition	Intention-behaviour profile formulation	Profile results
Fiala and Rhodes (2010)	337 undergraduate students in a 2-week prospective design	Moderate- or vigorous intensity physical activity performed at least 30 min per bout at least 4 × per week	Intention was measured with an open-ended frequency; behaviour was measured as an open-ended frequency (GLTEQ). Exercise was defined as moderate or vigorous intensity, at least 30+ min per bout. Profiles were dichotomized at 4+ times per week.	Non-intenders (37%), non-intenders exceeding intentions (4%), unsuccessful intenders (34%), and successful intenders (25%)
de Bruijn (2011)	328 undergraduate students in a 2-week prospective design	Vigorous exercise performed at least 20 min per bout at least 3 × per week	Intention was measured as the mean of two items on 7-point scales. Exercise behaviour was assessed with relevant items from the International Physical Activity Questionnaire. Profiles were dichotomized at meeting the exercise norm (3 times per week, at least 20 min per bout) or not. Intention was dichotomized at midscale values. Respondents at midscale values were excluded from the analysis.	Non-intenders (50.61%), non-intenders exceeding intentions (.01%), unsuccessful intenders (31.50%; mean bouts = 1.46, SD = 1.53), and successful intenders (17.88%)

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Table 1. (Continued)

Authors	Sample and design	Regular physical activity definition	Intention-behaviour profile formulation	Profile results
de Bruijn et al. (2012)	413 undergraduate students in a 2-week prospective design	Vigorous exercise performed at least 20 min per bout at least 3 × per week	Intention was measured as the mean of two items on 7-point scales. Exercise behaviour was assessed with relevant items from the International Physical Activity Questionnaire. Profiles were dichotomized at meeting the exercise norm (3 times per week, at least 20 min per bout) or not. Intention was dichotomized at midscale values. Respondents at midscale values were excluded from the analysis.	Non-intenders (32.85%), non-intenders exceeding intentions (.01%), unsuccessful intenders (37.46%; mean bouts = 1.54, SD = .87), and successful intenders (29.68%).
de Bruijn et al. (2009)	186 participants from a convenience sample of adults in a 1-month prospective design	Accumulating 60 min per week of vigorous -intensity physical activity Accumulating 150 min per week of moderate- or higher-intensity physical activity	Intention was measured with an open-ended frequency for moderate and vigorous PA separately. Physical activity was assessed with the International Physical Activity Questionnaire. Profiles were dichotomized at 150 min per week (moderate PA) and 60 min per week (vigorous PA).	Moderate PA Non-intenders (11.83%), non-intenders exceeding intentions (not reported), unsuccessful intenders (32.25%; mean bouts = 3.62, SD = 1.65), and successful intenders (55.91%) Vigorous PA Non-intenders (22.04%), non-intenders exceeding intentions (not reported), unsuccessful intenders (17.74%; mean bouts = .97, SD = 1.02), and successful intenders (60.22%)

Analysis

Quantitative evidence synthesis was performed. Effect sizes in the form of percentiles were calculated for proportions of group membership to each intention–behaviour profile. Meta-analysis procedures were based on random-effects models with correction for sampling error using the procedures recommended by Hunter and Schmidt (2004). Still, the heterogeneity across studies was also of interest, so both the Q -statistic and I -squared statistics from fixed-effects models were also included. Q -statistic tests the hypothesis that the observed variance in effect sizes is no greater than that expected by sampling error alone, while I -squared quantifies the dispersion. For interpretation, we considered I -squared values of 25, 50, and 75 as low, moderate, and high, respectively (Higgins, Thompson, Deeks, & Altman, 2003). Data were analysed using Comprehensive Meta-analysis-2 (Biostat, 2006).

Results

The prior review by Rhodes and Dickau (2012b) showed 2,865 records, and 248 relevant abstracts were subsequently obtained and assessed. From these, five papers met inclusion criteria (de Bruijn, de Groot, van den Putte, & Rhodes, 2009; Godin *et al.*, 1986; Rhodes & Plotnikoff, 2006; Rhodes, Plotnikoff, & Courneya, 2008; Rhodes *et al.*, 2003). The new literature search yielded 48 potentially relevant records. Of these, 31 abstracts and full-text reports were obtained and reviewed and two additional studies were identified (de Bruijn, 2011; Rhodes, de Bruijn, & Matheson, 2010). Authors from these studies were subsequently contacted to enquire about any additional research, and this yielded three additional studies (de Bruijn, Verkooijen, de Vries, & van den Putte, 2012; Fiala & Rhodes, 2010; Rhodes, Fiala, & Nasuti, 2012). Studies were screened to make sure all papers contained original data sets and were not multiple publications from the same data set. All studies contributed original data. Thus, 10 studies describing unique samples passed the eligibility criteria and were included (see Figure 1 based on QUOROM/PRISMA guidelines).

Study characteristics

The settings of the studies included universities ($k = 6$), community settings ($k = 2$), and the workplace ($k = 2$) (see Table 1). All samples included adults within employment ages (i.e., 19–64 years). Sample size ranged from 153 to 1,192. Participants were of either gender. Physical activity was assessed using the Godin Leisure Time Exercise Questionnaire (GLTEQ) (Godin & Shephard, 1985) or International Physical Activity Questionnaire (IPAQ) (Craig *et al.*, 2003). All of the studies were prospective in design, ranging from 2 weeks to 6 months between intention and physical activity assessments. The majority of the studies were conducted in Canada ($k = 7$), with the remaining being conducted in the Netherlands ($k = 3$).

Meta-analysis of intention–behaviour profiles

All 10 studies (see Table 1; total $N = 3,899$) yielded the information sufficient to be included in meta-analysis of the intention–behaviour profiles. Those non-intenders who subsequently did not engage in physical activity represented 21% (observed error variance = .06; sampling error = .00) of the samples, and non-intenders who subsequently performed physical activity comprised 2% (observed error variance = .00;

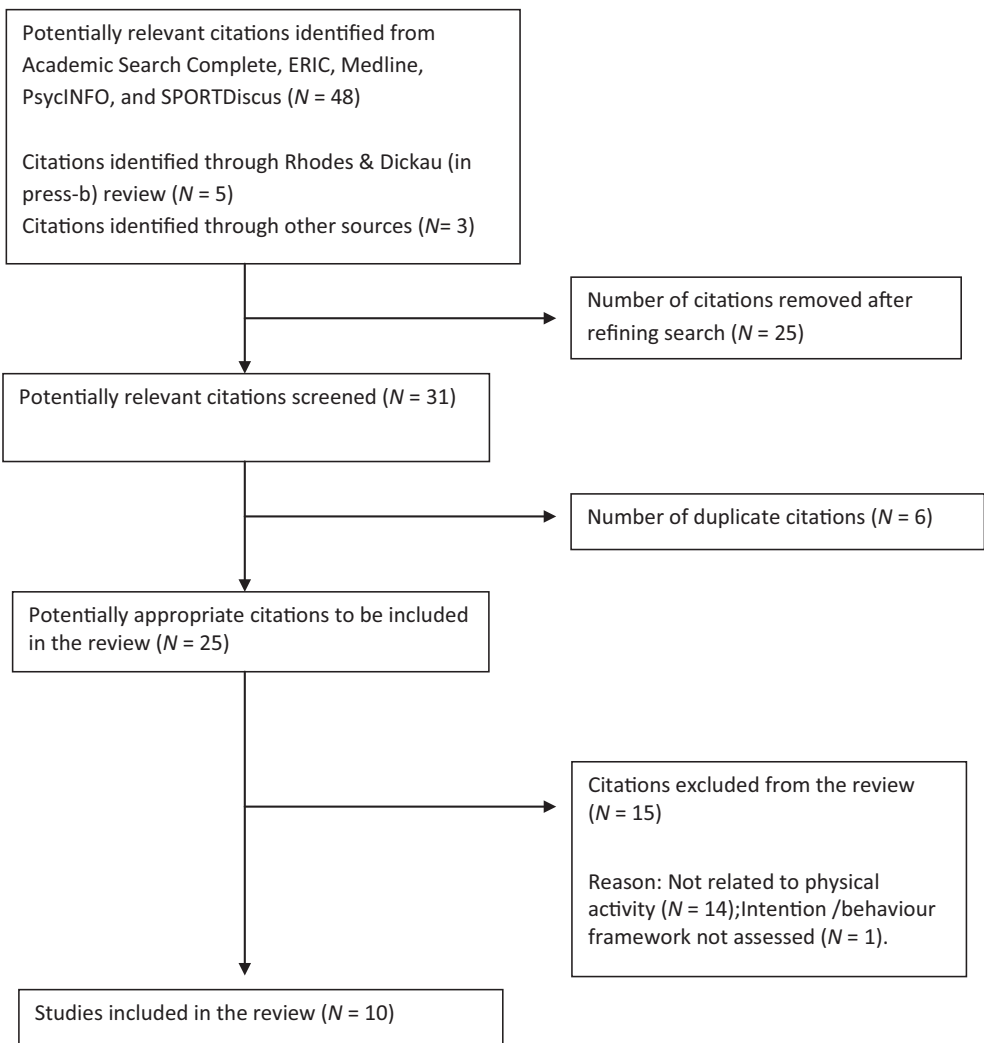


Figure 1. Literature search flow using QUOROM/PRISMA guidelines.

sampling error = .00). In terms of the action control profiles, those intenders who were not successful at following through with their exercise plans represented 36% of the sample (observed error variance = .12; sampling error = .00), while intenders who fulfilled their physical activity motivation represented 42% (observed error variance = .17; sampling error = .00). Considering the action control population specifically, the ratio of unsuccessful intenders to successful intenders shows that the intention-behaviour gap for physical activity represents 46% of the sample. The specific analysis for point-estimate heterogeneity showed that the non-intenders who subsequently performed physical activity [$Q(9) = .90, p = 1.00; I^2 = .00$] and intenders who were not successful at following through with exercise [$Q(9) = 15.65, p = .07; I^2 = 42.5$] had homogeneous findings/low spread across the 10 studies. By contrast, non-intenders who subsequently did not engage in physical activity [$Q(9) = 83.44, p = .00; I^2 = 89.21$] and intenders who

were subsequently active [$Q(9) = 55.94$, $p = .00$; $I^2 = 83.91$] had considerable heterogeneity and spread. Subsequent moderator analysis was not performed due to the small total number of studies.

Discussion

The intention-PA gap is a topic of considerable contemporary research, given that most of our models used to understand physical activity suggest that intention is the proximal antecedent of behavioural enactment. The purpose of this study was to quantify the intention-PA gap with a meta-analysis of the action control framework in order to appraise the magnitude of variation for intention at public health guidelines. The results should be useful as researchers develop new models and interventions to address intention-behaviour discordance in the physical activity domain.

Our random-effects meta-analysis of 10 studies that have employed the action control framework and 3,899 participants showed that nearly twice as many people fail to translate their intentions into physical activity (36%) than simply have no intention to be active (21%). This finding yields important information, because traditional intention-based theories suggest that changing intention is the primary intervention target (Fishbein & Ajzen, 2005). The results suggest that, while some people still require interventions changing intentions, much of the inactive population has positive intentions.

Perhaps more telling was the quantification of the intention-behaviour gap. Our meta-analysis showed that only 54% of intenders were able to successfully perform that intended behaviour. It is also important to note that only 2% of participants comprised non-intenders who were subsequently active. This is strong evidence that intention is a necessary, but often insufficient construct to produce behavioural enactment. The findings need to be considered within the context that two of the four action control quadrants (non-intenders who subsequently did not engage in physical activity and intenders who were subsequently active) showed heterogeneity among the 10 studies. Still, these findings clearly support the move towards models that focus on intention translation in interventions. Intention-behaviour discordance is not simply from high or low score deviations. People who intend to be physically active at conventional public health guidelines (e.g., Public Health Agency of Canada, 2002) show considerable deviation, which suggests the impact on typical interventions is potentially important. Indeed, subanalysis of available data showed that unsuccessful intenders were short of guidelines by 50% (1.5 bouts of the recommended three bouts per week) (see Table 1).

These findings support the contemporary line of enquiry for models that place emphasis on a volitional phase of post-motivational constructs that are self-regulatory in nature (Gollwitzer & Brandstatter, 1997; Schwarzer, 2008; Sniehotka, 2009). Alternatively, dual-process models that include motivational/regulatory constructs and habit/automaticity constructs would seem an important area of continued research. The initial evidence with these models supports the tenet that behavioural enactment may be from motivation and automation (Gardner *et al.*, 2011). It would seem these additional constructs may be needed to account for this intention-behaviour discordance.

Another area of research that may need attention is the measurement of motivation with traditional intention items. It may be that motivation is still present in the physical activity intention-behaviour gap, but intention is simply an insufficient measure to

capture this variance. The results from a prior review show that intention stability, considered the best measure of the strength of an intention, is the most robust moderator of the intention–behaviour gap (Rhodes & Dickau, 2012b). Sustained research on the measurement of motivation seems prudent. Clearly, a gap of 48% supports the possibility that both motivational and post-motivational domains could be relevant to the current intention–behaviour discordance.

The findings of this review need to be considered within the context of its limitations and possibilities for future research. Six of the 10 studies included undergraduate samples, so a wider range of sampling may be prudent. We saw no difference between community samples and undergraduate samples in terms of these results (see Table 1), but young people, older adults, and special populations are not represented. Second, an assessment of prediction time frame across the action control framework is needed. The studies reviewed ranged from 2-week to 6-month prospective design, and while the current sample is too small to examine any moderation by time frame, it may be an important future research question. Third, the cut-offs for physical activity intention and behaviour differ across some of the studies. This is partly due to the differences in guidelines across countries and the various interpretations of these guidelines. For example, Canadian guidelines (Health Canada, 2002) traditionally included a frequency and duration component, while many other international guidelines just include total minutes (World Health Organization, 2012). Guidelines also state two separate instructions for moderate and vigorous intensities. Still, there were no obvious differences across studies that seemed to align with the variations in guideline cut-offs, and a specific moderation analysis of action control by moderate/vigorous intensity yielded null findings (Fiala & Rhodes, 2010). Still, these differences could contribute to some heterogeneity among the studies. Fourth, the use of this framework has been limited to regions of Canada and the Netherlands. Replication at other regions would help establish the generalizability of these results. Finally, the action control framework has been applied almost exclusively to PA, yet it may have utility in many other health and social behaviours. Future research outside of the PA domain could also prove interesting.

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Appendix A: Search syntax

Narrowed by: Human, English language, all adult 19+ years, March 2010–September 2012.

Databases: Academic Search Complete, ERIC, MEDLINE, PsycINFO, and SPORTDiscus.

1. Intention behaviour and action control and physical activity
2. Intention behaviour and action control and exercise
3. Intention behaviour and inclined actor and physical activity
4. Intention behaviour and inclined actor and exercise
5. Intention behaviour and disinclined actor and physical activity
6. Intention behaviour and disinclined actor and exercise
7. Intention behaviour and disinclined abstainer and physical activity
8. Intention behaviour and disinclined abstainer and exercise
9. Intention behaviour and inclined abstainer and physical activity
10. Intention behaviour and inclined abstainer and exercise